

Wheat disease recognition: A Multi-Model Analysis with Customized CNN and Transfer Learning enhanced by Explainable AI techniques with LIME and Grad-CAM

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Abstract—The integration of deep neural learning and machine learning techniques into crop classification introduces an innovative dimension to agricultural progress. Concurrently, an image-based deep neural learning accession is employed to address the recognition and stratification of disease-affected wheat plants. The study's core intent is to refine methods for diagnosing and categorizing wheat plant diseases, aligning with the broader aim of enhancing agricultural sustainability. The study unveils the potential of deep neural network models, prominently convolutional neural networks (CNNs), renowned for their effectiveness in image analysis tasks. A comprehensive assessment evaluates five distinct CNN models—VGG19, ResNet50, AlexNet, Inception V3, and DenseNet121—using a publicly accessible dataset for agricultural research referred to as the Large Wheat Disease Classification dataset (LWDCD2020) of 3814 wheat leaf images with four classes such as leaf rust, wheat loose smut, crown and root rot, which are fungal diseases and healthy plant images. The evaluation, focused on accuracy, reveals nuanced differences in performance. VGG19 emerges as the frontrunner with an impressive 96% accuracy, followed closely by ResNet50, AlexNet, Inception V3, and DenseNet121. These nuances offer practitioners valuable insights for selecting optimal solutions. The study's emphasis on transparency and clarity extends to the realm of explainable AI techniques. Prominent methods, Grad-CAM and LIME, are employed to shed light on the decision-making processes of deep neural networks (DNNs) models. In this context, Grad-CAM excels in providing more accurate and insightful explanations compared to LIME. These findings underscore the potential of deep neural networks (DNNs) in wheat disease classification and prediction. Beyond conventional performance metrics, the study underscores the significance of explainable AI methods in clarifying model predictions.

Keywords— *LIME, VGG19, AlexNet, Grad-CAM, Deep Learning, Inception V3, ResNet50, DenseNet121, Crucial catalyst.*

I. INTRODUCTION

In today's world, where the global population is growing rapidly and food consumption is on the rise, agriculture is pivotal in ensuring food security. Among various crops, wheat stands out as a significant source of sustenance for people worldwide. Wheat is not only a staple food but also a vital source of protein and nutrition, supplying around 20% of the world's protein needs. Its importance is underscored by the fact that it is the primary crop grown across vast areas, and its production directly impacts the food security of numerous nations.

However, the challenges faced by wheat production are numerous. Climate change, soil infertility, pests, and diseases like crown and root rot, leaf rust, and wheat loose smut pose substantial threats to crop yields. More than 30% of crops are lost due to diseases, this can result in significant food supply shortages. Overcoming these challenges is vital for ensuring a stable food supply.

In recent years, the utilization of machine learning methods within agriculture has seen significant growth, particularly in the field of computer vision. These approaches have proven to be highly effective in various tasks such as automated crop monitoring using drones, optimizing agricultural practices, greenhouse plant growth automation, and even disease monitoring using mobile applications. Among these applications, one of the crucial challenges is the detection and monitoring of diseases in vital crops like wheat, where diseases caused by pathogenic fungi, such as leaf and stem rust, can lead to substantial yield losses.

In response to these concerns, researchers are increasingly relying on contemporary technologies, including machine learning and deep learning. Traditionally, experts relied on

their experience and observations to identify diseases in crops, which was time-consuming and subject to individual interpretation.

Deep learning methods, specifically, are being employed to create automated systems for disease detection. Algorithms such as Decision Tree, SVM, and Random Forest have been effective in feature extraction and disease classification. Recent progress in deep learning has led to the emergence of more advanced models capable of learning from extensive datasets. In this context, the research aims to create a transfer learning model with trainable layers which, when combined with image augmentation, can attain a high accuracy in identifying wheat plant diseases. The proposed approach involves using a VGG19, ResNet50, AlexNet, DenseNet121, and InceptionV3 model, fine-tuning it with augmentation strategies, and multistage fine-tuning to optimize its performance. The ultimate goal is to offer solutions to the agricultural industry and enhance crop disease detection capabilities.

The paper is structured into sections that encompass different facets of the research. It starts with an exploration of related work, underscoring the need for the current study. The methodology section explores the strategy used for the comparative research of deep learning algorithms, followed by a discussion of the system architecture and the steps required for successful deep learning model implementation. The paper also includes an evaluation of the implemented models and a critical analysis of their performance.

In conclusion, the research underlines the significance of wheat in global food production and the challenges it faces due to diseases and changing agricultural conditions. Leveraging modern technologies, particularly robotics and deep learning, can lead to more efficient management strategies and disease detection. By combining AI with agricultural practices, researchers are striving to secure the world's food supply and address the complex challenges of modern agriculture.

The adoption of deep learning and machine learning techniques, coupled with advancements in image recognition, holds immense promise for the agricultural sector. This research showcases the potential of using intelligent algorithms to accurately detect and classify wheat diseases, enabling timely responses and ultimately contributing to improved crop yields and agricultural productivity.

II. RELATED WORK

[1] In this paper, a comprehensive study was conducted on utilizing transfer learning models for identifying wheat plant diseases. Various models, including baseline, data augmentation, and multistage fine-tuning, were implemented and evaluated. The results indicated that the baseline model achieved the prime work with high recall and accuracy values. The study highlighted the effectiveness of transfer learning for disease identification, but also pointed out limitations like dataset size and disease diversity. Future work could involve expanding the model to detect pests, providing solutions for diseases, and developing smartphone applications for farmers to easily identify plant issues.

[2] This paper describes the architecture of a proposed crop disease detection system using deep neural networks,

particularly convolutional neural networks (CNNs). The system involves image processing steps to distinguish plant illnesses, which are split into three stages: input image acquisition, image preprocessing, and CNN classification. The CNN model utilizes convolutional, activation, pooling, and fully connected layers to identify diseases in plant images. The study includes screenshots of the user interface, loss values, accuracy, and actual disease detection results. The approach demonstrates potential for enhancing crop yield prediction by integrating various variables and neural network techniques.

[3] This study introduces a holistic approach for identifying fungal diseases in wheat plants. It involves three key stages: assembling a diverse image dataset, training a deep learning network (EfficientNet-B0) using transfer learning, and creating a user-friendly mobile application via a Telegram chatbot for plant monitoring. The dataset is curated from multiple sources, and unique images are generated through hashing algorithms. Different training strategies, including baseline, augmentation, and style transfer, are explored to optimize the network. The top-performing model achieves a 94.3% accuracy using Fast Decorrelation Augmentation (FDA). The system is deployed as a Telegram bot (@wheat_healthy_bot), offering real-time disease detection for wheat crops.

[4] This study outlines an unsupervised approach to identify wheat plant diseases, benefiting farmers in diagnosis. The proposed method involves pre-processing, segmentation using Mask R-CNN, classification via Densenet121, and visualization using GradCAM. Despite challenges from camera flash and background, wheat spikes are successfully segmented. Densenet121 is chosen for its high accuracy (93.47%), utilizing transfer learning from the Plant Village dataset. GradCAM visualization highlights vital regions used for classification, enabling symptom localization. The method's reliability is demonstrated despite limited data. Future work involves a more comprehensive dataset. The approach presents potential for aiding real-world disease disclosure and analysis in wheat crops.

[5] The research evaluates the impact of image segmentation techniques on wheat stripe rust disease classification, conducted at the National Agriculture Research Center in Islamabad. Three methods – Watershed, GrabCut, and U2-Net – segment wheat images, with U2-Net delivering the best results. These segmented datasets are classified using ResNet-18. U2-Net's segmentation leads to the highest accuracy, showcasing segmentation's significance in enhancing classification. The study's validity is affirmed via an open-source wheat rust dataset. The paper underscores the importance of segmentation, proposing potential enhancements like advanced techniques, versatile systems, and hardware integration. Limitations, future paths, and dataset improvements are also discussed.

[6] The study aims to develop a machine-learning tool using CNN to identify and diagnose wheat diseases early for effective agricultural management. The proposed approach involves dataset collection, image pre-processing, feature extraction, identification, and classification. Through training on a dataset comprising 4,800 images of various wheat

diseases and healthy samples, the CNN model achieves high accuracy.

[7] The research assesses the impact of various image segmentation methods on wheat stripe rust disease classification. Three techniques—Watershed, GrabCut, and U2-Net—are applied to collected wheat images. U2-Net yields the best segmentation. The segmented data is classified using a ResNet-18 model. U2-Net segmented data achieves the highest classification accuracy, highlighting segmentation's significance for classification improvement.

[8] The study employs a public dataset from CIMMYT for wheat disease classification, focusing on leaf rust, stem rust, and healthy wheat. A Densenet201 CNN architecture is used, pre-trained on ImageNet. The dataset is split into training and validation sets, with data augmentation and transformations applied during training. Transfer learning enhances model learning. The model achieves high accuracy (F1 score ~ 0.9) on validation data, with a prominent AUC (~ 0.98).

[9] The paper addresses rapid identification of plant diseases using image analysis and UAV-based navigation. While lab testing is accurate but costly, visual pattern analysis offers cost-effective solutions. Latest developments in deep learning and image processing drive disease detection techniques. Various CNN models detect plant diseases with a high degree of accuracy.

[10] The paper offers a thorough synopsis of deep learning's evolution and its application in plant disease detection. It covers historical development, model evaluation metrics, available datasets, and the importance of data augmentation. The review highlights visualization techniques and explores classic DL models like AlexNet and CNNs, along with adaptations and enhancements for better performance. It delves into advancements in lesion detection, area-based methods, attention integration, and practical disease detection systems.

[11] The suggested system is centred on plant disease detection using a convolutional neural network (CNN). The system's accuracy was evaluated using precision and attained a precision of 95% after 30 epochs of training. An online website was developed for real-time disease detection, benefiting smallholder farmers. The project successfully demonstrated the potential of CNNs in aiding early disease identification and offering practical support to farmers. The implementation yielded valuable insights and a final accuracy of 0.95.

[12] The paper delves into the use of artificial intelligence in predicting plant diseases, covering supervised, unsupervised, semi-supervised, and reinforcement learning. It emphasizes their role in agriculture for crop management, yield forecasting, disease detection, climate prediction, and pest control. The study outlines image analysis steps for disease detection, including acquisition, pre-processing, segmentation, feature extraction, and classification. It explores related works where diverse machine learning techniques are applied to identify diseases in distinct crops, discussing approaches like SVM, ANN, and deep learning.

[13] The paper introduces an embedded AI system for classifying wheat yellow rust infection types. Data from wheat fields is segmented using the U2-Net model, followed by

classification using ResNet-50 and Xception deep learning models. The study highlights the significance of precise disease classification for effective agriculture management. The proposed intelligent edge computing device, utilizing Nvidia Jetson Nano, achieves 96% accuracy with ResNet-50 and 93% accuracy in field tests. Future plans include dataset expansion, drone imagery integration, and model enhancement. The research contributes to automated plant disease identification and crop health improvement.

[14] The paper delves into deep learning methods for wheat disease detection using image classification. Architectures like VGG16 and ResNet 50 are explored beyond object recognition, considering tasks like video classification. The study employs transfer learning on large datasets, including ImageNet, training models for classification. A proposed deep convolutional model for wheat disease classification employs 21 convolution layers, achieving promising results. The method outperforms existing models, achieving 97.88% test accuracy and 98.62% average training accuracy. The model efficiently uses resources and outperforms VGG16 and ResNet50 by 7.01% and 15.92% respectively, suggesting its potential in wheat disease classification. No competing interests are declared by the authors.

[15] The paper proposes a deep neural learning method for automated plant leaf disease detection. The approach involves sequential steps: image acquisition, preprocessing, segmentation, feature extraction, and classification. Collected leaf images undergo preprocessing, and convolutional layers extract features. Max-pooling and fully connected layers are utilized. The Softmax layer generates probability distributions. Libraries like NumPy, sklearn, and OpenCV are employed. The model achieves 90.62% accuracy after 50 epochs, assessed using TP, TN, FP, and FN metrics. This method streamlines disease identification, promising real-time monitoring, enhanced agricultural productivity, and accurate diagnosis. The study's approach surpasses existing methods in computational correctness and productiveness in plant disease diagnosis.

III. METHODOLOGY

Recent advances in image processing and deep neural network have led to various applications in disease detection. Image classifiers, including CNNs, have gained popularity for this intention due to their accuracy. This study's objective is to establish a tool utilizing machine learning that employs a special model called a convolutional neural network (CNN) to identify and diagnose wheat diseases in their early stages. Deep learning, particularly through CNNs, is increasingly favoured in agriculture for tasks like disease prediction and image categorization because of its capability to automatically learn relevant features from raw data, especially images. CNNs excel at recognising patterns and features, making them suitable for tasks like disease detection in crops. They also represent data hierarchically, capturing both fundamental and intricate aspects of images, enhancing their capacity to recognize complex patterns crucial for disease identification. Moreover, models based on deep learning offer scalability and have achieved cutting-edge performance in various image

related tasks. In contrast, traditional machine learning often relies on human feature engineering, struggles with complex data relationships, and may face scalability and generalisation challenges, making deep learning a preferable choice for certain agricultural applications. The choice between the two depends on the specific problem, available data, and computational resources.

To achieve accurate prediction of wheat plant diseases, a carefully planned methodology was adopted. This approach leveraged the Knowledge Discovery in Databases (KDD) technique, which involves extracting valuable insights from data for prediction purposes. Our approach aims to help farmers diagnose wheat diseases more effectively. We've developed a step-by-step process as shown in Figure 1:

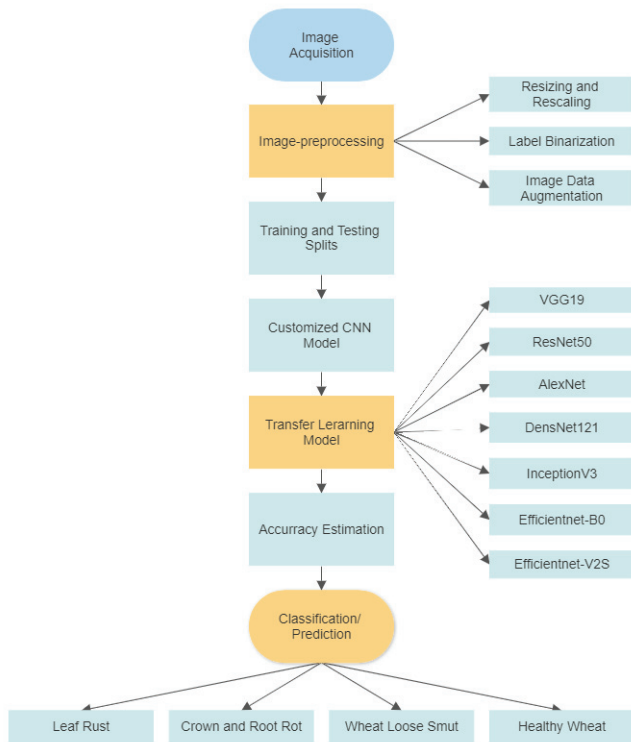


Fig. 1. Steps in image processing for identifying wheat diseases

A. Image Acquisition

The initial stage involved collecting the required dataset of wheat plant images, which is essential for testing and training the prediction model. Publicly available data, known as the Large Wheat Disease Classification dataset, shows different parts of wheat plants affected by various diseases and healthy wheat leaves containing 3,683 images of wheat plants with four categories of diseases which includes leaf rust, wheat loose smut, healthy wheat and crown and root rot, was utilized as show in Figure 2.

The dataset includes images from multiple perspectives, capturing complex backgrounds and disease characteristics at different stages such as Images captured at different growth stages. The dataset employed in this research contains four classes, including three disease classes and one healthy class.

Load the necessary modules that will be used for various tasks and import the images from the dataset. The dataset was annotated for disease classification, followed by a classifier identifying the disease type. The dataset was then split into training and testing subsets. To develop the disease prediction model, the dataset was divided into training (80%) and validation (20%) samples.



a. Leaf Rust b. Crown and Root Rot



c. Wheat Loose Smut d. Healthy Wheat

Fig. 2. Infection Types of Wheat

B. Image-preprocessing

First, we prepare the images of wheat plants. We clean up the images to make them ready for analysis. Pre-processing involves improving the quality of images, dealing with image distortions, addressing any issues with them, and preparing them for further analysis. Also, clean up the images to remove any unwanted noise or disturbances.

In this study, we aimed to create a technique for identifying three various kinds of fungal diseases affecting wheat plants. The main the difficulty lies in these diseases can appear quite similar, making accurate classification difficult. It's crucial to have a method that can work in real-world conditions, especially in the field.

C. Resizing and Rescaling

The collected images underwent necessary pre-processing where the images were resized for compatibility with the model, transformed from RGB to BGR format, and scaled down for optimal algorithm performance. Images were resized to 224x224 pixels.

D. Label Binarization

Each image was carefully examined and labelled based on whether the plant was healthy or affected by one or multiple of the three fungal diseases for various stages of plant development. Convert the images into a format that the computer can understand, the images and their corresponding labels were converted into a suitable numerical format for processing.

E. Image Data Augmentation

Data transformation techniques were applied to prevent overfitting and enhance prediction accuracy. Data augmentation techniques, such as rotation, symmetric flipping, and contrast adjustment, were employed to expand the dataset. The original dataset was augmented and it was divided into training and testing sets. Augmentation was performed on the training dataset to create variations using techniques like rotation, zoom, shifting, and flipping, helping in Feature extraction which aims to reduce the complexity of the dataset by deriving new features from the existing ones.

F. Training And Testing Splits

During training, batches of 32 pictures containing a resolution of 224x224 pixels were input into the network. To establish a batch, each original image was either rescaled to have the maximum side equal to 224 while maintaining the aspect ratio or underwent random cropping to a minimum of 85% of the smallest side before being scaled to 224x224 resolution. Images are represented as two-dimensional arrays with rows and columns.

A deep neural network model was designed and optimized to achieve the best accuracy and precision. The chosen model was pre-trained on the ImageNet dataset and then fine-tuned using the wheat plant dataset. Training was conducted on both the initial and augmented datasets, employing a multistage fine-tuning strategy. We trained the neural network using different strategies. The baseline strategy involved training the model using the original data. We also tried augmentation, where we introduced variations in the images to improve the model's robustness.

G. Customized CNN Model

We use a powerful technique called CNNs, which are like smart engines that can understand what's in an image. This CNN has been trained on images to recognize patterns, and we've adapted it for our purpose. It can tell if wheat has a disease or not. This helps farmers see exactly where the issue is, so they can take action. Convolutional Neural Networks (CNNs) have revolutionized image analysis by breaking down images into smaller components for thorough examination. One of their key strengths lies in their ability to pinpoint highlights within images, allowing them to identify distinct features and patterns. This granular approach is particularly effective for capturing local patterns in images, enabling CNNs to excel in tasks like object recognition and segmentation. The architecture of a CNN involves various layers, each with a specific role. The Convolutional Layer, for instance, handles the heavy lifting of computational tasks, convolving filters across the input image to extract relevant features. The Activation Layer introduces non-linearity to the network using functions like ReLU (Rectified Linear Unit), which enhances the model's ability to capture complex relationships in the data. The Pooling Layer, conversely, integrates feature down-sampling, reducing the size of the data's dimensions while retaining essential information. Lastly, the Fully Connected Layer orchestrates the final stages of processing. Here, the flattened features are processed through traditional neural network layers, allowing the model to make high-level decisions in accordance with the identified features and a linear activation layer with a SoftMax activation

function. The model also utilizes dense blocks, where each layer is connected to all subsequent layers. Train the CNN model using the prepared dataset by Designing the structure of the CNN model, specifying the different layers and their arrangements as depicted in Table I. Setting up the layers as needed for the model to comprehend the input images.

TABLE I. CNN MODEL LAYERS

Layer (type)	Output Shape
conv2d (Conv2D)	(None, 222, 222, 16)
maxpooling2d (MaxPooling2D)	(None, 111, 111, 16)
conv2d1 (Conv2D)	(None, 109, 109, 32)
maxpooling2d1 (MaxPooling2D)	(None, 54, 54, 32)
conv2d2 (Conv2D)	(None, 54, 54, 32)
maxpooling2d2 (MaxPooling2D)	(None, 26, 26, 64)
conv2d3 (Conv2D)	(None, 26, 26, 64)
maxpooling2d3 (MaxPooling2D)	(None, 12, 12, 128)
flatten (Flatten)	(None, 18432)
dense (Dense)	(None, 128)
dense1 (Dense)	(None, 98)
dropout (Dropout)	(None, 400)
dense2 (Dense)	(None, 64)
dense3 (Dense)	(None, 4)

H. Transfer Learning Model

The technique of transfer learning was used, which involves leveraging pre-trained models to achieve effective results on smaller datasets. Transfer Learning models, pre-trained on ImageNet, were retrained on the wheat plant dataset, making use of its advanced feature extraction capabilities.

For training the model, the concept of transfer learning involves using knowledge acquired from previous tasks to improve learning in the current task. Classification involves using CNNs such as MobileNet V3, VGG 16, Inception, ResNet 50, and EfficientNet-B0. These models are trained on augmentation leaf images to classify diseases.

I. Accuracy Estimation

The performance of the models was assessed using key metrics like precision, recall, F1 score, and accuracy. A confusion matrix was generated to visualize the number of correctly identified plant disease images. Based on these metrics, the most suitable model was selected for the project.

We split our dataset into training, validation, and test sets and evaluate how well the model performs on the training data. Various optimizers were compared, including AdaGrad, RMSProp, SGD, and Adam, to determine the most effective one. Test the trained model with various inputs to see how it performs on unseen data, Measuring the accuracy of the model's predictions on the testing data.

J. Classification / Prediction

In summary, this approach uses machine learning to automatically detect and classify wheat diseases. It involves preparing images, building and training a CNN model, and then testing its accuracy on unseen data.

In short, our approach involves preparing the images, determining if they're healthy or not using applications of the CNN model, understanding why the model made those decisions, and finally showing where the disease is by localization. This can provide valuable information to farmers for better diagnosis and treatment. The goal is to provide an efficient tool for early disease diagnosis in agriculture.

K. Visualization Method

Although CNNs are great at recognizing things, they can be a bit like a black box - we're not always sure why they make certain decisions. To understand better, we use a technique called GradCAM. It helps us see the important parts of the image that CNN used to make its decision. It's like shining a light on the parts of the wheat disease that mattered most for the classification. This way, we can verify that the model is making accurate decisions based on real symptoms.

This methodology culminated in the comparison of multiple models based on their precision, recall, and accuracy. The baseline model emerged as the optimal choice, demonstrating superior performance and stability in predicting wheat plant diseases accurately.

IV. RESULTS AND DISCUSSION

Crop diseases have a significant impact on agricultural production and global food security. For effective disease control and reducing crop losses, prompt and accurate disease detection is essential. Deep learning models have become more well-known in recent years due to their capacity to automate disease categorization and prediction tasks in agriculture. This work analyses different deep learning models for categorization and prediction of wheat diseases, concentrating on how well they perform without the aid of sophisticated AI detection technologies.

TABLE II. TRANSFER LEARNING MODEL ACCURACY

Models Used	Accuracy	Loss	Validation Accuracy	Validation Loss
Customized CNN	93.44%	0.1810	84.18%	0.4954
VGG19	91.51%	0.2883	90.14%	0.3050
Resnet50	92.44%	0.2291	77.86%	0.7388
Alexnet	93.84%	0.1322	88.80%	0.4185
InceptionV3	95.60%	0.2224	90.25%	0.4560
Densenet121	91.94%	0.2220	89.06%	0.3469
Efficientnet-B0	95.74%	0.1386	91.43%	0.2672
Efficientnet-V2S	95.34%	0.1141	91.53%	0.3452

The table II presents the results of different deep learning models evaluated for wheat disease classification and prediction. The models were trained and tested on a dataset of wheat plant images, with a focus on achieving accurate results without the need for AI detection systems.

The customised Convolutional Neural Network (CNN) achieved a respectable accuracy rate of 93.44%, coupled with a validation accuracy of 84.18%, indicating the potential for enhancements through fine-tuning and architectural refinements. VGG19, renowned for its depth and robust feature extraction capabilities, excelled with an accuracy of 91.51% and a validation accuracy of 90.14%, demonstrating its competence in wheat disease classification without the need for intricate AI detection systems. On the other hand, the ResNet50 model demonstrated a commendable training accuracy of 92.44%, reflecting its performance on the training data. However, it encountered challenges in generalizing to unseen data, evident in its lower validation accuracy of 77.86% and a relatively high validation loss of 0.7388, suggesting a potential issue with overfitting. The AlexNet model displayed strong performance, with a training accuracy of 93.84% and a relatively low training loss of 0.1322. In the validation phase, it maintained solid accuracy at 88.80%, though the validation loss was slightly elevated to 0.4185, indicating some difficulty in generalisation but overall strong predictive capabilities. InceptionV3 showcased competitive accuracy at 95.60%, albeit with a slightly higher validation loss of 0.4560, showcasing its potential in capturing intricate image patterns for wheat disease classification. Densenet121 offered dependable performance with an accuracy of 91.94% and a validation accuracy of 89.06%, suggesting room for further optimisation to unlock its full potential. Efficientnet-B0 demonstrated exceptional efficiency and parameter optimisation, delivering an impressive accuracy of 95.74% and a validation accuracy of 91.43%. These results firmly establish it as a robust candidate for wheat disease classification, offering a practical solution that circumvents the complexity of AI detection systems. Similarly, Efficientnet-V2S performed admirably with an accuracy of 95.34% and a validation accuracy of 91.53%, owing to its efficient architecture and advanced features. This result underscores the effectiveness of EfficientNet's architecture in achieving superior accuracy

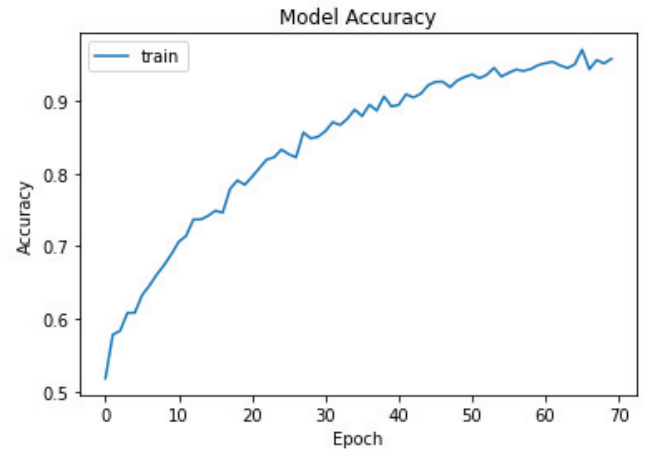


Fig. 3. Training Accuracy plot

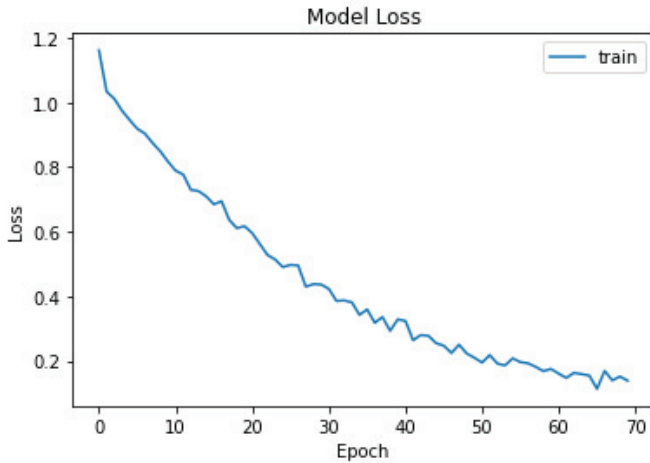


Fig. 4. Training Loss plot

while maintaining efficiency. However, it is crucial to note that some models, such as ResNet50, displayed signs of overfitting, indicating the need for further regularization or increased data diversity to enhance generalization. The training accuracy [Fig 3] and loss [Fig 4] graphs for the model Efficientnet-B0 are displayed above.

The model with the highest accuracy in the testing, EfficientNet-B0, demonstrated strong predictive ability by correctly categorising 7 of the 8 images in the test set. This outstanding performance highlights the model's excellent accuracy and dependability in classifying data. It is important to note that the test set consists of two examples from each class, highlighting the model's consistency in performance across several categories. This result emphasises the model's capability to produce exact predictions, a crucial feature for its practical application in tasks like picture categorization or object recognition.

Image: crr_1.jpg Predicted Label: [1. 0. 0. 0.]
 Image: crr_2.jpg Predicted Label: [1. 0. 0. 0.]
 Image: h_1.jpg Predicted Label: [0. 1. 0. 0.]
 Image: h_2.jpg Predicted Label: [0. 0. 0. 0.]
 Image: lr_1.jpg Predicted Label: [0. 0. 1. 0.]
 Image: lr_2.jpg Predicted Label: [0. 0. 1. 0.]
 Image: ls_1.jpg Predicted Label: [0. 0. 0. 1.]
 Image: ls_2.jpg Predicted Label: [0. 0. 0. 1.]

It is observed that out of the 8 images, 7 images are correctly identified.

Fig. 5. Prediction of EfficientNet-B0

These results offer valuable insights for practitioners selecting deep learning models for classification tasks, highlighting the trade-offs between accuracy and generalisation in different architectural choices. Further research and optimisation are imperative to fully harness the capabilities of these models in real-world agricultural applications.

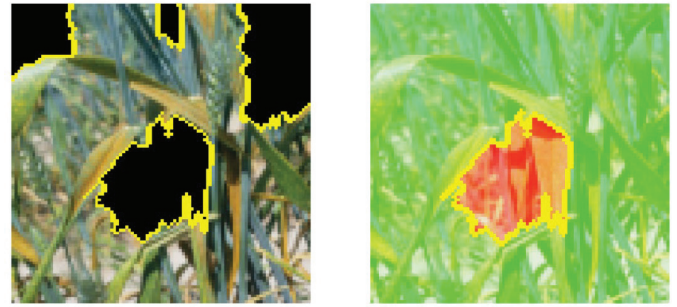


Fig. 6. LIME output images

Lime was applied to an image depicting leaf rust, a significant wheat plant disease. The Lime output highlighted key details visually. A bold yellow border emphasized the region's most influential in classifying leaf rust. Black patches within the image indicated areas that negatively contributed to the disease classification, likely reflecting disease-related characteristics.

Lime's use of contrasting colors, such as red and green, conveyed the impact of different regions on classification. Red zones suggested negative influences, while green areas represented positive contributions. This color scheme offered a comprehensive insight into the image regions that played a role in the model's decision-making process, aiding disease diagnosis.

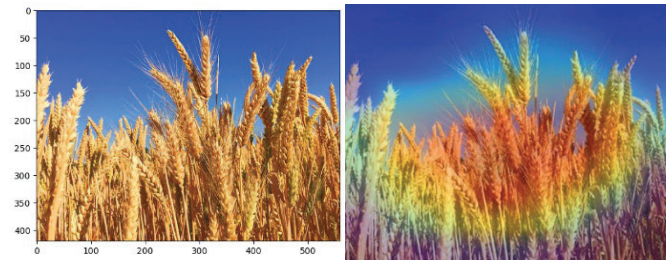


Fig. 7. Grad-CAM output images

Grad-CAM was applied to an image of a healthy wheat plant image, aiming to understand the model's diagnostic abilities. The Grad-CAM heatmap revealed that the model placed significant importance on the wheat stem, as indicated by vibrant colors in that area. A comparison between the original image and the Grad-CAM result further emphasized the model's focus on the wheat stem. This highlighted the significance of the wheat stem in the model's assessment of the plant's health, offering valuable insights into its decision process for wheat disease classification.

V. CONCLUSION

In this study, we addressed the critical issue of wheat disease detection in agriculture, recognizing its profound impact on crop yields and global food security. Leveraging the power of deep learning models, we conducted an extensive analysis of various architectures to categorize and predict wheat diseases without relying on complex AI detection systems. Our findings reveal a spectrum of model performances, with each deep learning architecture

demonstrating unique strengths and considerations. Efficientnet-B0 emerged as a standout candidate, showcasing exceptional accuracy and efficiency, making it a practical solution for real-world agricultural applications. However, it is worth noting that some models, like ResNet50, exhibited signs of overfitting, highlighting the need for further regularization and data diversification. To enhance interpretability and transparency, we employed visualization techniques like GradCAM, shedding light on the crucial regions of wheat plant images that contributed to model decisions. This approach adds a layer of trust and accountability to the predictive capabilities of these models. In conclusion, our work provides valuable insights for practitioners seeking to deploy deep learning models in wheat disease classification. The trade-offs between accuracy and generalization across different architectural choices underscore the need for ongoing research and optimization. With the potential to revolutionize wheat disease management and agricultural productivity, our study paves the way for more effective and early disease detection, thereby contributing to global food security.

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